

DATA DICTIONARY – SHIFAA

Comprehensive Database Schema & Entity Relationship Specification

1. DOCUMENT SCOPE

Business/database role. The design uses User for account data and separate Patient, Doctor, and Caregiver profile entities.

2. ERD NOTATION & RULES

Participation is represented by a double line; Partial participation by a single line. Cardinality is expressed as 1, N, or 0..1.

3. RELATIONSHIP SUMMARY

#	Entity A	Relationship	Entity B	Cardinality	Participation
1	User	is	Patient	1:0..1	User P / Patient T
2	User	is	Doctor	1:0..1	User P / Doctor T
3	User	is	Caregiver	1:0..1	User P / Caregiver T
4	Hospital	has	Department	1:N	Hospital P / Department T
5	Patient	has	PatientCaregiver	1:N	Patient P / PatientCaregiver T
6	Caregiver	has	PatientCaregiver	1:N	Caregiver P / PatientCaregiver T
7	Patient	books	Appointment	1:N	Patient P / Appointment T
8	Doctor	has	Appointment	1:N	Doctor P / Appointment T
9	Appointment	results in	Visit	1:0..1	Appointment P / Visit T
10	Visit	has	Assessment	1:N	Visit P / Assessment T
11	Visit	has	Diagnosis	1:N	Visit P / Diagnosis T
12	Visit	has	Medical Test	1:N	Visit P / Medical Test T
13	Doctor	requests	Medical Test	1:N	Doctor P / MedicalTest T
14	Medical Test	has	TestResult	1:0..1	MedicalTest P / TestResult T
15	Visit	has	MedicalImage	1:N	Visit P / MedicalImage T
16	Doctor	reviews	MedicalImage	1:N	Doctor P / MedicalImage T
17	Patient	has	Medical Document	1:N	Patient P / Medical Document T
18	User	uploads	Medical Document	1:N	User P / Medical Document T
19	Patient	has	VitalSign	1:N	Patient P / VitalSign T
20	Patient	has	TreatmentPlan	1:N	Patient P / TreatmentPlan T
21	Doctor	creates	TreatmentPlan	1:N	Doctor P / TreatmentPlan T
22	Diagnosis	based on	TreatmentPlan	1:N	Diagnosis P / TreatmentPlan P
23	TreatmentPlan	has	Prescription	1:N	TreatmentPlan P / Prescription T
24	Prescription	contains	PrescriptionItem	1:N	Prescription T / PrescriptionItem T
25	Medicine	used in	PrescriptionItem	1:N	Medicine P / PrescriptionItem T
26	PrescriptionItem	tracks	Medicine Tracking	1:N	PrescriptionItem P / Medicine Tracking T
27	Patient	records	Medicine Tracking	1:N	Patient P / Medicine Tracking T
28	Patient	owns	Device	1:N	Patient P / Device T
29	Device	produces	Device Reading	1:N	Device P / Device Reading T
30	Patient	triggers	Alert	1:N	Patient P / Alert T

#	Entity A	Relationship	Entity B	Cardinality	Participation
31	User	receives	Notification	1:N	User P / Notification T
32	Alert	generates	Notification	1:N	Alert P / Notification P
33	Appointment	has	Online Visit	1:0..1	Appointment P / OnlineVisit T
34	Chat	has	ChatParticipant	1:N	Chat T / ChatParticipant T
35	User	joins	ChatParticipant	1:N	User P / ChatParticipant T
36	Chat	has	Message	1:N	Chat P / Message T
37	User	sends	Message	1:N	User P / Message T
38	Patient	grants	Consent	1:N	Patient P / Consent T
39	User	receives access	Consent	1:N	User P / Consent T
40	User	creates	AuditLog	1:N	User P / AuditLog T

4. ENTITY SPECIFICATIONS

1. User

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
UserID	INT / UUID	PK	No	Unique account identifier.	Primary identifier; referenced by profile and security-related records.
Email	VARCHAR		No	Login email.	Authentication / contact address.
PasswordHash	VARCHAR		No	Secure password hash.	Stores a hash, never plain-text password.
Role	VARCHAR / ENUM		No	System role.	Patient, Doctor, Caregiver, Admin.
Status	VARCHAR / ENUM		No	Account state.	Active, Inactive, Suspended, etc.
CreatedAt	DATETIME		No	Account creation timestamp.	Lifecycle tracking.
LastLoginAt	DATETIME		Yes	Last successful login.	Nullable before first login.

2. Patient

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
PatientID	INT / UUID	PK	No	Patient profile identifier.	Identifies the patient.
UserID	INT / UUID	FK	No	User.UserID reference.	Links patient profile to account.
FirstName	VARCHAR		No	Patient first name.	Personal identity.
LastName	VARCHAR		No	Patient last name.	Family name.
MedicalRecordNo	VARCHAR		No	Medical record number.	Business identifier for medical record.
DateOfBirth	DATE		No	Date of birth.	Patient demographic data.
Gender	VARCHAR / ENUM		No	Gender.	Demographic data.
BloodType	VARCHAR / ENUM		Yes	Blood group.	May be unknown.
Address	VARCHAR		Yes	Patient address.	Contact / location information.
EmergencyContact	VARCHAR		Yes	Emergency contact name.	Emergency communication.

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
EmergencyPhone	VARCHAR		Yes	Emergency contact phone.	Emergency communication.
RegistrationDate	DATE		No	Registration date.	Patient registration tracking.
Status	VARCHAR / ENUM		No	Patient record state.	Active, Inactive, etc.

3. Doctor

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
DoctorID	INT / UUID	PK	No	Doctor profile identifier.	Identifies the doctor.
UserID	INT / UUID	FK	No	User.UserID reference.	Links doctor profile to account.
FirstName	VARCHAR		No	Doctor first name.	Personal identity.
LastName	VARCHAR		No	Doctor last name.	Family name.
LicenseNumber	VARCHAR		No	Medical license number.	Professional identifier.
Specialization	VARCHAR		No	Medical specialization.	Doctor specialty.
YearsOfExperience	INT		Yes	Years of experience.	Professional profile data.
Bio	TEXT		Yes	Professional biography.	Optional profile description.
Status	VARCHAR / ENUM		No	Doctor record state.	Active / inactive status.

4. Caregiver

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
CaregiverID	INT / UUID	PK	No	Caregiver profile identifier.	Identifies the caregiver.
UserID	INT / UUID	FK	No	User.UserID reference.	Links caregiver profile to account.
FirstName	VARCHAR		No	Caregiver first name.	Personal identity.
LastName	VARCHAR		No	Caregiver last name.	Family name.
Relationship	VARCHAR		No	Relationship to patient.	Parent, spouse, sibling, etc.
Phone	VARCHAR		No	Caregiver phone.	Communication.
Address	VARCHAR		Yes	Caregiver address.	Optional location data.
Status	VARCHAR / ENUM		No	Caregiver record state.	Active / inactive status.

5. Hospital

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
HospitalID	INT / UUID	PK	No	Hospital identifier.	Identifies facility.
Name	VARCHAR		No	Hospital name.	Facility name.
Type	VARCHAR / ENUM		No	Facility type.	Classifies hospital.
Address	VARCHAR		No	Hospital address.	Facility location.
Phone	VARCHAR		Yes	Hospital phone.	Facility contact.
Email	VARCHAR		Yes	Hospital email.	Facility contact.
Status	VARCHAR / ENUM		No	Hospital status.	Active / inactive.

6. Department

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
DepartmentID	INT / UUID	PK	No	Department identifier.	Identifies department.

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
HospitalID	INT / UUID	FK	No	Hospital.HospitalID reference.	Department's hospital.
Name	VARCHAR		No	Department name.	Department label.
Description	TEXT		Yes	Department description.	Purpose / function.
Status	VARCHAR / ENUM		No	Department status.	Active / inactive.

7. PatientCaregiver

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
PatientCaregiverID	INT / UUID	PK	No	Association identifier.	Identifies patient-caregiver relationship.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Patient side of association.
CaregiverID	INT / UUID	FK	No	Caregiver.CaregiverID reference.	Caregiver side of association.
PermissionLevel	VARCHAR / ENUM		No	Caregiver access level.	Controls permitted access.
StartDate	DATE		No	Relationship/access start.	Beginning of association.
EndDate	DATE		Yes	Relationship/access end.	Null while ongoing.
Status	VARCHAR / ENUM		No	Association status.	Active, Ended, Revoked.

8. Appointment

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
AppointmentID	INT / UUID	PK	No	Appointment identifier.	Identifies scheduled appointment.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Appointment patient.
DoctorID	INT / UUID	FK	No	Doctor.DoctorID reference.	Appointment doctor.
Date	DATE		No	Appointment date.	Scheduled date.
StartTime	TIME		No	Start time.	Beginning of slot.
EndTime	TIME		No	End time.	End of slot.
Type	VARCHAR / ENUM		No	Appointment type.	Classifies appointment.
Status	VARCHAR / ENUM		No	Appointment state.	Scheduled, Completed, Cancelled, No-Show.
Reason	TEXT		Yes	Reason for appointment.	Booking purpose.
Notes	TEXT		Yes	Additional notes.	Optional context.

9. Visit

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
VisitID	INT / UUID	PK	No	Visit identifier.	Identifies clinical encounter.
AppointmentID	INT / UUID	FK	No	Appointment.AppointmentID reference.	Links encounter to appointment.
Date	DATE		No	Visit date.	Actual encounter date.
Type	VARCHAR / ENUM		No	Visit type.	Encounter classification.
MainComplaint	TEXT		Yes	Main patient complaint.	Primary concern.
Notes	TEXT		Yes	Clinical notes.	Encounter notes.
Status	VARCHAR / ENUM		No	Visit status.	Encounter state.

10. Assessment

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
AssessmentID	INT / UUID	PK	No	Assessment identifier.	Identifies assessment.
VisitID	INT / UUID	FK	No	Visit.VisitID reference.	Assessment's visit.
Date	DATE		No	Assessment date.	Date performed.
Type	VARCHAR / ENUM		No	Assessment type.	Assessment classification.
Symptoms	TEXT		Yes	Reported symptoms.	Patient-reported symptoms.
Observations	TEXT		Yes	Clinical observations.	Observed findings.
RiskLevel	VARCHAR / ENUM		Yes	Risk level.	Low, Medium, High, etc.
Notes	TEXT		Yes	Assessment notes.	Additional information.

11. Diagnosis

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
DiagnosisID	INT / UUID	PK	No	Diagnosis identifier.	Identifies diagnosis.
VisitID	INT / UUID	FK	No	Visit.VisitID reference.	Diagnosis's visit.
Name	VARCHAR		No	Diagnosis name.	Human-readable diagnosis.
Code	VARCHAR		Yes	Classification code.	Standard code if available.
Description	TEXT		Yes	Diagnosis description.	Clinical details.
Date	DATE		No	Diagnosis date.	Date recorded.
Severity	VARCHAR / ENUM		Yes	Diagnosis severity.	Severity classification.
Status	VARCHAR / ENUM		No	Diagnosis status.	Active/resolved etc.

12. MedicalDocument

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
DocumentID	INT / UUID	PK	No	Document identifier.	Identifies medical file.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Document owner.
UploadedBy	INT / UUID	FK	No	User.UserID reference.	Uploader.
Type	VARCHAR / ENUM		No	Document type.	Report, Prescription, Record, etc.
Name	VARCHAR		No	Document name.	Display name.
FilePath	VARCHAR		No	Storage path/key.	Location of stored file.
UploadDate	DATETIME		No	Upload timestamp.	Tracks upload.
Description	TEXT		Yes	Document description.	Optional details.
Status	VARCHAR / ENUM		No	Document status.	Availability/validity.

13. Medical Test

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
TestID	INT / UUID	PK	No	Test identifier.	Identifies requested test.
VisitID	INT / UUID	FK	No	Visit.VisitID reference.	Related visit.
DoctorID	INT / UUID	FK	No	Doctor.DoctorID reference.	Requesting doctor.
Name	VARCHAR		No	Test name.	Medical test name.
Type	VARCHAR / ENUM		No	Test type.	Test classification.

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
RequestedDate	DATE		No	Request date.	Date ordered.
Status	VARCHAR / ENUM		No	Test status.	Requested, In Progress, Completed.
Notes	TEXT		Yes	Test notes.	Instructions/context.

14. TestResult

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
ResultID	INT / UUID	PK	No	Result identifier.	Identifies stored result.
TestID	INT / UUID	FK	No	Medical Test.TestID reference.	Related test.
Date	DATE		No	Result date.	Availability date.
Value	VARCHAR		No	Observed value.	Measured result.
Unit	VARCHAR		Yes	Measurement unit.	Result unit.
NormalRange	VARCHAR		Yes	Reference range.	Expected range.
Interpretation	TEXT		Yes	Result interpretation.	Clinical interpretation.
Notes	TEXT		Yes	Result notes.	Additional comments.

15. MedicalImage

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
ImageID	INT / UUID	PK	No	Image identifier.	Identifies medical image.
VisitID	INT / UUID	FK	No	Visit.VisitID reference.	Related visit.
DoctorID	INT / UUID	FK	No	Doctor.DoctorID reference.	Reviewing/associated doctor.
Type	VARCHAR / ENUM		No	Imaging type.	X-Ray, CT, MRI, Ultrasound.
Date	DATE		No	Image date.	Imaging date.
FilePath	VARCHAR		No	Storage path/key.	Image location.
Report	TEXT		Yes	Imaging report.	Associated report.
Status	VARCHAR / ENUM		No	Image status.	Availability/review state.

16. VitalSign

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
VitalSignID	INT / UUID	PK	No	Vital-sign identifier.	Identifies one measurement.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Measured patient.
Type	VARCHAR / ENUM		No	Vital-sign type.	Heart rate, BP, temperature, etc.
Value	VARCHAR		No	Measured value.	Recorded measurement.
Unit	VARCHAR		Yes	Measurement unit.	bpm, °C, mmHg, etc.
DateTime	DATETIME		No	Measurement timestamp.	When measured.
Source	VARCHAR / ENUM		No	Measurement source.	Manual or device source.

17. TreatmentPlan

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
TreatmentPlanID	INT / UUID	PK	No	Treatment-plan identifier.	Identifies treatment plan.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Plan patient.
DoctorID	INT / UUID	FK	No	Doctor.DoctorID reference.	Plan creator.
DiagnosisID	INT / UUID	FK	Yes	Diagnosis.DiagnosisID reference.	Optional related diagnosis.
StartDate	DATE		No	Treatment start.	Beginning of plan.
EndDate	DATE		Yes	Treatment end.	Null if ongoing.
Description	TEXT		No	Treatment description.	Planned treatment.
Goals	TEXT		Yes	Treatment goals.	Expected outcomes.
Status	VARCHAR / ENUM		No	Treatment status.	Active, Completed, Cancelled.
Notes	TEXT		Yes	Treatment notes.	Additional clinical notes.

18. Prescription

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
PrescriptionID	INT / UUID	PK	No	Prescription identifier.	Identifies prescription.
TreatmentPlanID	INT / UUID	FK	No	TreatmentPlan.TreatmentPlanID reference.	Related treatment plan.
Date	DATE		No	Prescription date.	Issue date.
Status	VARCHAR / ENUM		No	Prescription status.	Active/completed/cancelled.
Instructions	TEXT		Yes	General instructions.	Overall medication directions.
Notes	TEXT		Yes	Prescription notes.	Additional information.

19. Medicine

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
MedicineID	INT / UUID	PK	No	Medicine identifier.	Catalog identifier.
Name	VARCHAR		No	Medicine name.	Brand/human-readable name.
GenericName	VARCHAR		Yes	Generic name.	Active ingredient/generic.
Form	VARCHAR / ENUM		Yes	Dosage form.	Tablet, capsule, syrup, etc.
Strength	VARCHAR		Yes	Medicine strength.	500 mg, 10 mg/ml, etc.
Description	TEXT		Yes	Medicine description.	Additional information.
Status	VARCHAR / ENUM		No	Medicine status.	Active/inactive catalog state.

20. PrescriptionItem

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
PrescriptionItemID	INT / UUID	PK	No	Prescription-item identifier.	One medicine entry in a prescription.
PrescriptionID	INT / UUID	FK	No	Prescription.PrescriptionID reference.	Parent prescription.

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
MedicineID	INT / UUID	FK	No	Medicine.MedicineID reference.	Prescribed medicine.
Dose	VARCHAR		No	Dose per administration.	Amount per dose.
Frequency	VARCHAR		No	Administration frequency.	Once daily, every 8 hours, etc.
Route	VARCHAR / ENUM		Yes	Administration route.	Oral, IV, topical, etc.
Duration	VARCHAR		Yes	Treatment duration.	How long to take.
Instructions	TEXT		Yes	Medicine-specific instructions.	Detailed directions.

21. Medicine Tracking

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
TrackingID	INT / UUID	PK	No	Tracking identifier.	Identifies adherence event.
PrescriptionItemID	INT / UUID	FK	No	PrescriptionItem.PrescriptionItemID reference.	Tracked medicine.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Patient taking medicine.
ScheduledTime	DATETIME		No	Scheduled dose time.	Expected dose time.
TakenTime	DATETIME		Yes	Actual dose time.	Null if not taken.
Status	VARCHAR / ENUM		No	Adherence status.	Taken, Missed, Skipped.
Notes	TEXT		Yes	Tracking notes.	Additional adherence information.

22. Device

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
DeviceID	INT / UUID	PK	No	Device identifier.	Connected health device.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Device owner.
Type	VARCHAR / ENUM		No	Device type.	Smartwatch, glucose monitor, etc.
Name	VARCHAR		No	Device name.	Display name.
Manufacturer	VARCHAR		Yes	Manufacturer.	Device maker.
SerialNumber	VARCHAR		Yes	Serial number.	Hardware identifier.
ConnectedDate	DATETIME		No	Connection timestamp.	When connected.
Status	VARCHAR / ENUM		No	Device status.	Connected / disconnected.
LastSync	DATETIME		Yes	Last sync timestamp.	Null before first sync.

23. Device Reading

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
ReadingID	INT / UUID	PK	No	Reading identifier.	One device-generated reading.
DeviceID	INT / UUID	FK	No	Device.DeviceID reference.	Reading source.
Type	VARCHAR / ENUM		No	Reading type.	Heart rate, glucose, steps, etc.
Value	VARCHAR		No	Reading value.	Measured value.

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
Unit	VARCHAR		Yes	Measurement unit.	bpm, mg/dL, etc.
DateTime	DATETIME		No	Reading timestamp.	When produced.

24. Alert

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
AlertID	INT / UUID	PK	No	Alert identifier.	Health/system alert.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Associated patient.
Type	VARCHAR / ENUM		No	Alert type.	Alert classification.
Severity	VARCHAR / ENUM		No	Alert severity.	Low, Medium, High, Critical.
Title	VARCHAR		No	Alert title.	Short heading.
Description	TEXT		No	Alert description.	Condition/details.
CreatedAt	DATETIME		No	Creation timestamp.	When generated.
Status	VARCHAR / ENUM		No	Alert status.	New, Acknowledged, Resolved.
ResolvedAt	DATETIME		Yes	Resolution timestamp.	Null until resolved.

25. Notification

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
NotificationID	INT / UUID	PK	No	Notification identifier.	Identifies notification.
UserID	INT / UUID	FK	No	User.UserID reference.	Recipient.
AlertID	INT / UUID	FK	Yes	Alert.AlertID reference.	Optional source alert.
Type	VARCHAR / ENUM		No	Notification type.	Alert, appointment reminder, etc.
Title	VARCHAR		No	Notification title.	Short heading.
Message	TEXT		No	Notification message.	Delivered content.
CreatedAt	DATETIME		No	Creation timestamp.	When generated.
ReadAt	DATETIME		Yes	Read timestamp.	Null until read.
Status	VARCHAR / ENUM		No	Notification status.	Unread, Read, Archived.

26. Online Visit

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
OnlineVisitID	INT / UUID	PK	No	Online-visit identifier.	Telemedicine session.
AppointmentID	INT / UUID	FK	No	Appointment.AppointmentID reference.	Related appointment.
StartTime	DATETIME		No	Session start.	Beginning of online session.
EndTime	DATETIME		Yes	Session end.	Null until ended.
Type	VARCHAR / ENUM		No	Online visit type.	Telemedicine classification.
Status	VARCHAR / ENUM		No	Session status.	Scheduled, Active, Completed, Cancelled.
MeetingLink	VARCHAR		No	Meeting link/identifier.	Access to session.
Notes	TEXT		Yes	Session notes.	Additional information.

27. Chat

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
ChatID	INT / UUID	PK	No	Chat identifier.	Identifies conversation.
CreatedAt	DATETIME		No	Creation timestamp.	When chat was created.
Type	VARCHAR / ENUM		No	Chat type.	Conversation classification.
Status	VARCHAR / ENUM		No	Chat status.	Active / inactive.

28. ChatParticipant

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
ChatParticipantID	INT / UUID	PK	No	Participation identifier.	User membership in chat.
ChatID	INT / UUID	FK	No	Chat.ChatID reference.	Chat.
UserID	INT / UUID	FK	No	User.UserID reference.	Participant.
JoinedAt	DATETIME		No	Join timestamp.	When participant joined.
Status	VARCHAR / ENUM		No	Participant status.	Active / Left.

29. Message

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
MessageID	INT / UUID	PK	No	Message identifier.	Identifies message.
ChatID	INT / UUID	FK	No	Chat.ChatID reference.	Conversation.
SenderID	INT / UUID	FK	No	User.UserID reference.	Message sender.
MessageText	TEXT		No	Message content.	Text sent.
SentAt	DATETIME		No	Send timestamp.	When sent.
ReadAt	DATETIME		Yes	Read timestamp.	Null until read.
Status	VARCHAR / ENUM		No	Message status.	Sent, Delivered, Read, Deleted.

30. Consent

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
ConsentID	INT / UUID	PK	No	Consent identifier.	Permission record.
PatientID	INT / UUID	FK	No	Patient.PatientID reference.	Patient granting access.
GrantedToUserID	INT / UUID	FK	No	User.UserID reference.	User receiving access.
Type	VARCHAR / ENUM		No	Consent/access type.	Defines permitted data / action.
GrantedAt	DATETIME		No	Grant timestamp.	When access was granted.
ExpiresAt	DATETIME		Yes	Expiration timestamp.	Null if non-expiring.
Status	VARCHAR / ENUM		No	Consent status.	Active, Expired, Revoked.
RevokedAt	DATETIME		Yes	Revocation timestamp.	Null if not revoked.

31. AuditLog

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
LogID	INT / UUID	PK	No	Audit-log identifier.	Identifies audit event.
UserID	INT / UUID	FK	No	User.UserID reference.	User performing action.
Action	VARCHAR		No	Performed action.	CREATE, UPDATE, DELETE, LOGIN, etc.

Attribute	Suggested Type	Key	Nullable	Definition	Database / Business Role
EntityName	VARCHAR		No	Affected entity/table.	Business object affected.
EntityID	VARCHAR		No	Affected record ID.	Identifier of affected record.
DateTime	DATETIME		No	Action timestamp.	When action occurred.
IPAddress	VARCHAR		Yes	Source IP address.	Security/audit context.
Description	TEXT		Yes	Audit details.	Additional event context.

5. FINAL DESIGN NOTES

- **User/Profile Separation:** User credentials and core account details are isolated from profile-specific entities (Patient, Doctor, Caregiver).
- **Optional Foreign Keys:** `TreatmentPlan.DiagnosisID` and `Notification.AlertID` are nullable because the related record is not mandatory in every case.
- **Optional One-to-One:** `Appointment-Visit` and `Appointment-OnlineVisit` use 1:0..1 in this proposed design.
- **Test Results:** `MedicalTest-TestResult` is modeled as 1:0..1. If the implementation requires multiple individual lab measurements per test, it should be changed to 1:N.
- **Implementation:** Suggested data types are implementation-oriented and should be adapted to the selected DBMS during relational mapping.